

Human-centric Computing and Information Sciences

October 2025 | Volume 15



KCIA

Korea Computer
Industry Association

www.hcisjournal.com

A Comprehensive Review on Non-terrestrial Network Technologies in 3GPP Standards

Heejae Park¹, Kihyun Seol¹, Seungyeop Song¹, Yerin Lee¹, WonSeok Choi², Kyung-Mo Sung², Tri-Hai Nguyen^{3,*}, and Laihyuk Park^{1,*}

Abstract

Non-terrestrial networks (NTNs) are pivotal in addressing the limitations of terrestrial networks, particularly by extending connectivity to remote and underserved areas. This paper comprehensively reviews NTN technologies, focusing on architectural advancements and their integration with terrestrial networks. A chronological analysis of the 3rd Generation Partnership Project (3GPP) standards evolution from Release 15 to Release 19 highlights critical milestones in NTN standardization and deployment. Future research directions are identified, including regenerative payloads, advanced mobility management approaches, uplink/downlink enhancements, and the integration of reduced-capability devices to improve NTN efficiency and scalability. Additionally, transformative applications of NTNs across industries are explored, with discussions focusing on real-world use cases, operational challenges, and potential solutions. Open issues such as NTN-based open radio access networks, NTN-based artificial intelligence-enhanced radio access networks, Starlink integration, and network switching strategies are presented, offering insights into achieving seamless connectivity and operational efficiency. By addressing these aspects, the paper lays a robust foundation for advancing NTN technologies and their integration into next-generation wireless communication systems.

Keywords

3GPP, Non-terrestrial Networks, Terrestrial Networks, Architecture, Service Scenarios

1. Introduction

The integration of non-terrestrial networks (NTNs), which encompass spaceborne platforms in various earth orbits and airborne platforms, into the communication infrastructure is revolutionizing global connectivity [1]. In the context of 5G, NTNs address critical challenges such as expanding connectivity to underserved areas, ensuring service continuity during disasters, and supporting the Internet of Things (IoT) and autonomous systems [2]. NTNs complement terrestrial networks by providing seamless coverage, improving network resilience, and enabling high-speed, low-latency communication in remote regions. As 6G development progresses, NTNs are expected to play an even more transformative role.

* This is an Open Access article distributed under the terms of the Creative Commons Attribution Non-Commercial License (<http://creativecommons.org/licenses/by-nc/3.0/>) which permits unrestricted non-commercial use, distribution, and reproduction in any medium, provided the original work is properly cited.

*Corresponding Author: Tri-Hai Nguyen (hai.nguyentri@vlu.edu.vn), Laihyuk Park (lhpark@seoultech.ac.kr)

¹Department of Computer Science and Engineering, Seoul National University of Science and Technology, Seoul, Korea

²Department of Intelligent Network, Telecommunications Technology Association, Seongnam, Korea

³Faculty of Information Technology, Van Lang School of Technology, Van Lang University, Ho Chi Minh City, Vietnam.

Key 6G use cases, such as holographic communications, global autonomous mobility, and ultra-reliable low-latency communication, demand capabilities that NTN are uniquely positioned to deliver. The integration of advanced NTN technologies, including regenerative payloads, inter-satellite links, and artificial intelligence-enhanced radio access networks (AI-RANs), will facilitate unprecedented levels of connectivity and adaptability. By bridging the gap between terrestrial and extraterrestrial networks, NTNs are essential to realize the vision of ubiquitous, intelligent, and sustainable global communication in 6G. The significance of NTNs is amplified with the progression of 5G and 6G technologies, including the integrated access and backhaul (IAB), millimeter-wave (mmWave) and sub-terahertz (sub-THz) bands; network slicing; integrated sensing and communication (ISAC); and advanced beamforming and beam management techniques. In the context of IAB, NTNs play a crucial role in ensuring reliable backhaul connections and facilitating seamless data transmissions between remote access points and core networks [3]. Furthermore, NTNs contribute significantly to the utilization and expansion of the mmWave and sub-THz frequency bands, enabling high-capacity high-speed connectivity in areas that would otherwise remain underserved [4]. Integrating NTNs within 5G's network slicing framework enables the dynamic allocation of network resources, offering customized connectivity solutions for diverse applications such as IoT, autonomous systems, and emergency services [5]. In addition, NTNs enhance the potential of ISAC by supporting simultaneous communication and environmental sensing across vast regions, a capability that is vital for emerging applications such as real-time environmental monitoring, autonomous navigation, and disaster management [6]. Finally, NTNs improve beamforming and beam management techniques, ensuring reliable communication in complex and challenging environments. These advances enable high-performance, scalable services for both stationary and mobile users [7].

The 3rd Generation Partnership Project (3GPP) has been instrumental in standardizing NTN technologies, ensuring their seamless integration with the global cellular ecosystem. Since Release 15, NTNs have been a key focus in global strategies to bridge the connectivity divide, with normative requirements formally introduced in Release 17 [8, 9]. This milestone has enabled the integration of satellite and other non-terrestrial components with terrestrial networks, ensuring seamless service continuity and enhancing mobile communication capabilities worldwide [10]. Several studies have surveyed advancements in NTN technologies. For example, one paper [11] reviews 3GPP NTN features, state-of-the-art advancements, open issues, and future research directions, emphasizing NTN's role in advancing 5G and beyond networks. Other authors [12] explore cellular connected unmanned aerial vehicles (UAVs) in 5G and beyond, addressing challenges, solutions, and emerging technologies such as NTNs, artificial intelligence (AI), and open radio access networks (O-RANs). Similarly, another study [13] examines the standardization, architectures, and PHY/MAC procedures for NTNs in 5G, highlighting challenges such as Doppler shifts and large delays while proposing solutions for effective integration. Other surveys [14, 15] provide insights into the evolution of 3GPP Releases 15–18. Recent discussions in relation to 3GPP underscore key NTN technologies essential for expanding mobile network capabilities and addressing traditional terrestrial network challenges. These advancements provide valuable insights for developers and researchers, fostering innovation and improving next-generation mobile communication systems. Compared to prior research, this review paper is both timely and essential as NTNs transition from conceptual frameworks to practical, real-world deployments. It provides a comprehensive overview of recent advancements in NTN technologies, with a focus on 3GPP Releases 15–19. As the governing body for technical specifications, 3GPP manages all relevant documentation, including historical records and the latest versions of its standards. These documents are accessible through its official portal (<https://portal.3gpp.org/>), which offers direct access to technical specifications. Relevant details for this paper were retrieved using the portal's "Specification" tab, leveraging search terms such as "5G" AND "NTN OR Non-terrestrial network" AND "UAV OR Uncrewed aerial vehicle" AND "UAS OR Uncrewed aerial system OR Unmanned aerial systems" AND "Satellite OR Satellite access." The main contributions of this paper are as follows:

- We provide a comprehensive analysis of architectural advancements that enable the seamless integration of NTNs with terrestrial networks. We also discuss how NTNs facilitate the deployment

of advanced 5G/6G technologies, including reconfigurable intelligent surfaces (RISs), rate-splitting multiple access (RSMA), and grant-free random access, positioning these innovations as pivotal enablers for 6G.

- We present a detailed examination of the evolution of 3GPP standards, emphasizing their foundational role in supporting the deployment and operation of NTN. Additionally, we identify key future focus areas such as regenerative payloads, advanced mobility management, uplink and downlink enhancements, and the integration of reduced-capability (RedCap) devices. These developments are critical for improving the efficiency and scalability of NTNs.
- We explore the transformative impact of NTNs across diverse industries by addressing their scalability, inherent limitations, and real-world applications. Furthermore, we analyze various open issues, including NTN-based O-RAN, NTN-based AI-RANs, Starlink integration, and NTN switching strategies, which are essential for achieving seamless connectivity and operational efficiency in next-generation communication systems.

The remainder of this paper is organized as follows. A literature review is conducted in Section 2. Section 3 discusses the NTN communication architecture and its components. Section 4 reviews the 3GPP standards that underpin the NTN functionalities. Section 5 explores the service scenarios facilitated by NTNs. Finally, Section 6 concludes the paper.

2. Related Work

This section reviews existing literature on NTNs, summarized in Table 1 [16–30], encompassing UAV-assisted wireless communication, high-altitude platform stations (HAPS), and satellite communication networks.

One study [16] reviews UAV placement optimization strategies for 5G and beyond, analyzing centralized, semi-centralized, and distributed decision-making approaches. It focuses on optimization goals such as coverage, throughput, and latency while addressing challenges such as energy constraints, interference, and backhaul reliability. Future research directions are also identified, emphasizing the need for robust and scalable solutions. Another study [17] explores the evolution of UAV cellular communications from 5G to 6G, discussing advancements in technologies such as massive multiple-input multiple-output (MIMO), THz communications, and RISs. It identifies challenges, including interference and mobility management while highlighting future opportunities for integrating UAVs into non-terrestrial and cell-free networks. Similarly, other authors [18] propose a mixed-integer linear programming framework for the interference-aware trajectory planning of UAVs as flying base stations. This approach achieves up to a 21% throughput improvement with minimal additional completion time, demonstrating scalability and real-time applicability. With regard to disaster management, a three-stage stochastic model integrating 5G and UAVs for disaster preparedness, response, and recovery was presented [19]. The goal was to optimize service provision and network management under uncertainty, maximizing services while minimizing costs. Additional research [20] employs deep reinforcement learning to optimize UAV base station deployment for integrated access and backhaul in mission-critical scenarios, improving user throughput and ensuring reliable temporary connectivity.

For HAPS, a genetic algorithm-based two-step optimization method was introduced to enhance spectral efficiency and coverage in non-uniform user distributions [21]. The approach utilizes search space reduction and co-evolution techniques to improve the capacity and reduce the computational complexity. Meanwhile, other work [22] investigates the role of HAPS in 6G networks, emphasizing their potential for backhauling, dynamic spot beams, and computing capabilities. The study highlights challenges such as power constraints and integration complexity and proposes solutions to optimize performance outcomes in urban and rural scenarios. Additional work [23] analyzes the performance of HAPS-assisted hybrid radio frequency/free space optics (RF/FSO) systems, deriving outage probability and bit error rate expressions and proposing a power allocation scheme for energy-efficient operation under varying channel conditions. In a related vein, a cloud-enabled HAPS system that integrates

terrestrial and aerial networks was proposed, demonstrating improvements in coverage, fairness, and power management through advanced optimization techniques [24].

Table 1. Summary of prior research on NTN

Study	Summary
Elnabty et al. [16]	Examined UAV placement optimization strategies for UAV-assisted wireless communication in 5G and beyond, focusing on centralized, semi-centralized, and distributed approaches while addressing energy constraints and interference challenges.
Geraci et al. [17]	Analyzed the evolution of UAV cellular communications from 5G to 6G, emphasizing advancements such as massive MIMO and RIS, while discussing integration challenges and future opportunities.
Lee & Friderikos [18]	Proposed an interference-aware UAV trajectory planning framework using mixed integer linear programming, demonstrating significant throughput improvements with minimal delays.
Colajanni et al. [19]	Developed a three-stage stochastic model integrating 5G and UAVs for disaster management, optimizing service provision and network management under uncertain conditions.
Zhang et al. [20]	Utilized deep reinforcement learning to optimize UAV base station deployment, ensuring improved user throughput and reliable connectivity in dynamic and mission-critical environments.
Shibata et al. [21]	Proposed a genetic algorithm-based optimization method for HAPS cell configurations, enhancing spectral efficiency and coverage in non-uniform user distribution scenarios.
Abbasi et al. [22]	Investigated the role of HAPS in 6G networks, highlighting capabilities such as dynamic backhauling and computing, while addressing power limitations and integration complexity.
Deka et al. [23]	Evaluated HAPS-assisted hybrid RF/FSO systems, deriving performance metrics such as outage probability and bit error rates, and proposing power allocation schemes to optimize communication under real-world constraints.
Alghamdi et al. [24]	Proposed a cloud-enabled HAPS system for digital equity, leveraging advanced user scheduling and beamforming techniques to improve coverage, fairness, and resource management.
Jiang et al. [25]	Examined the application of game theory in satellite networks, addressing resource management challenges through cooperative and non-cooperative models for integration with 5G/6G systems.
Jiang [26]	Examined software-defined satellite networks, focusing on scalability, resource allocation, and seamless integration with terrestrial systems, while identifying key challenges and solutions.
Ding et al. [27]	Introduced a dynamic optimization framework for LEO satellite-assisted mobile edge computing, minimizing energy consumption and efficiently managing task offloading and resource allocation.
Duong et al. [28]	Explored machine-learning-based multi-beam optimization techniques for 6G satellite-terrestrial networks, addressing challenges related to latency, energy efficiency, and disaster recovery.
Cui et al. [29]	Investigated hybrid GEO-LEO satellite networks for IoT applications, employing collaborative computing to optimize task offloading, resource allocation, and latency reduction operations.
Voicu et al. [30]	Analyzed handover strategies for non-geostationary constellations to balance spectral efficiency, delay, and handover rates.

For satellite communication networks, one paper [25] surveys game theory applications, addressing resource limitations, management complexity, and strategic behavior. Cooperative and non-cooperative game models are explored for tasks including resource allocation, security, and task offloading, with insights into their integration with B5G/6G. Similarly, other work in this area [26] reviews software-defined satellite networks, focusing on scalability, resource allocation, and integration with terrestrial networks. Key challenges and opportunities, such as controller placement and virtualization, are discussed to enhance satellite network efficiency and flexibility. A dynamic optimization framework for dense LEO satellite-terrestrial mobile edge computing was also introduced, balancing energy consumption and delay through Lyapunov optimization and advanced algorithms [27]. It addresses task offloading, satellite association, and MIMO precoding. Similarly, another study [28] proposes machine-learning-based multi-beam optimization for 6G integrated satellite-terrestrial networks, tackling latency and energy efficiency challenges and highlighting applications in disaster relief and IoT. Hybrid GEO-LEO satellite networks

for IoT were also investigated, with deep reinforcement learning used to optimize task offloading and resource allocation [29]. The study highlights collaborative computing between satellites and ground stations to reduce latency and improve efficiency. Finally, a paper [30] evaluates handover strategies for non-geostationary constellations, comparing methods such as closest satellite, max visibility, and carrier-to-interference-and-noise ratio-based handovers. Their results indicate that optimal strategies depend on the constellation design, balancing spectral efficiency, delay, and handover rates.

3. NTN Communication Architecture

This section explores the foundational elements of NTN platforms, their reliance on 5G new radio (NR)-based architectures, and open issues.

3.1 Non-terrestrial Network Platforms

NTNs can be deployed through various platform types, broadly classified into spaceborne and airborne platforms. This classification, depicted in Fig. 1, is based on parameters such as the altitude, orbit type, and beam footprint size. Spaceborne platforms consist of three main categories:

- 1) Geostationary earth orbiting (GEO) satellites: Positioned approximately 35,786 km above the earth’s equator, GEO satellites maintain a fixed position relative to the earth. They provide extensive coverage with footprints ranging from 200 to 3,500 km.
- 2) Medium-earth orbiting (MEO) satellites: Orbiting at altitudes between 7,000 and 25,000 km, MEO satellites serve mid-range coverage needs, with footprints spanning 100 to 1,000 km.
- 3) Low-earth orbiting (LEO) satellites: Operating at altitudes between 300 and 1,500 km, LEO satellites offer smaller footprints, ranging from 100 to 1,000 km, and are well suited for low-latency communication.

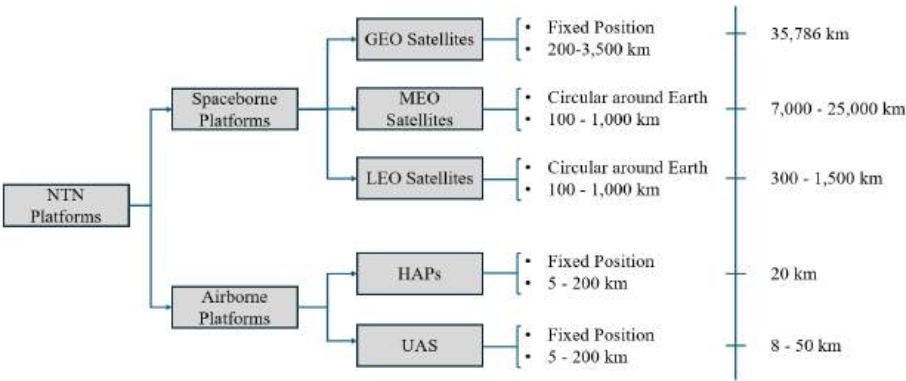


Fig. 1. NTN platform taxonomy.

Airborne platforms can be classified into two main types:

- 1) Unmanned aircraft system (UAS) platforms: UAS platforms operate at altitudes ranging from 8 to 50 km, supporting localized connectivity with footprints between 5 and 200 km.
- 2) High-altitude platforms (HAPs): Positioned at around 20 km, HAPs provide extended coverage with footprints comparable to the UAS platform, ranging from 5 to 200 km.

Table 2 highlights key differences between NTNs and terrestrial networks across various features [31, 32]. In particular, NTNs provide unparalleled global coverage, extending up to 3,500 km, which makes them indispensable for connecting remote and underserved areas, whereas terrestrial networks are constrained to regional coverage ranges of up to 100 km. Although communication rates are comparable—up to 150 Mbps for NTNs versus over 100 Mbps for terrestrial networks—NTNs face challenges such as higher latency (exceeding 500 ms for GEO satellites), significant propagation path loss (190 dB), and

pronounced Doppler shifts (up to 48 kHz for LEO satellites). In contrast, terrestrial networks exhibit lower latency (under 40 ms), reduced path loss (138 dB for 100 km), and minimal Doppler shifts (around 1 kHz in high-speed scenarios). While NTN incur higher infrastructure costs due to sophisticated satellite systems, they excel in reliability during disasters and accessibility in challenging environments. Terrestrial networks, with their cost-effective ground infrastructure, offer superior handover management and are optimized for long-term deployments. By providing flexible deployment options, NTNs address unique communication needs and, when integrated with terrestrial systems, create a complementary framework that bridges diverse connectivity gaps.

Table 2. Comparison of NTNs and terrestrial networks [31, 32]

Feature	NTNs	Terrestrial networks
Coverage area	Global coverage (up to 3,500 km for GEO satellites)	Regional coverage (up to 100 km)
Communication rates	Up to 150 Mbps (e.g., Starlink)	Typically exceeds 100 Mbps
Latency	High latency (e.g., over 500 ms for GEO satellites)	Low latency (e.g., below 40 ms)
Propagation path loss	Approximately 190 dB (GEO satellite at 2 GHz carrier frequency)	Approximately 138 dB (100 km cell at 2 GHz carrier frequency)
Doppler shift	Significant (e.g., around 48 kHz for LEO satellites at 600 km altitude)	Minimal (e.g., around 1 kHz for high-speed trains at 2 GHz)
Infrastructure cost	Very high due to satellite deployment	Lower due to reliance on ground-based infrastructure
Reliability	Highly reliable in disaster scenarios	Vulnerable to infrastructure damage during disasters
Accessibility	Provides connectivity in remote and underserved areas	Limited in remote or challenging terrains
Mobility support/ Handovers	Seamless mobility with periodic handovers (NGSO satellites)	Frequent handovers for mobile users
Network deployment	Supports both temporary and long-term deployments	Primarily long-term deployments

3.2 5G NR-based NTN Architecture

Integrating NTNs with the 5G NR framework is vital for ensuring seamless service and interoperability. Spaceborne and airborne platforms can be classified into two primary categories, differentiated by the specific type of payload they are designed to transport.

1) Transparent-payload-based NTN platforms: These platforms operate by receiving signals from the ground, amplifying them, and retransmitting them without altering the original signal [33]. Transparent payloads are simpler in design and cost effective, making them suitable for basic NTN services. However, their processing capabilities are limited to fundamental functions such as frequency conversion, amplification, and filtering. As a result, they offer a simpler and more cost-effective solution for basic NTN services.

2) Regenerative-payload-based NTN platforms: These platforms demodulate received signals to recover the baseband data, allowing for enhanced signal processing and improved quality. Regenerative payloads enable advanced functionalities, such as inter-satellite links, dynamic routing, and flexible gateway selection. Regenerative-payload-based NTN platforms incorporate onboard processing capabilities, allowing the satellite to go beyond simple relaying by demodulating, decoding, and possibly re-encoding signals before forwarding them. While more complex and costly than transparent payloads, they are better suited for sophisticated NTN applications, including those that require high-speed data transmission and

low-latency communication [34].

In addition to NTN platforms, NTN rely on several key components to enable communication [35]:

- NTN gateway: This infrastructure serves as a bridge between the NTN platform and the terrestrial data network.
- User equipment (UE): These are the devices such as smartphones, IoT devices, or specialized terminals that utilize NTN services.
- Service link: This link establishes the connection between the NTN platform and the UE.
- Feeder link: This link connects the NTN platform to the NTN gateway.

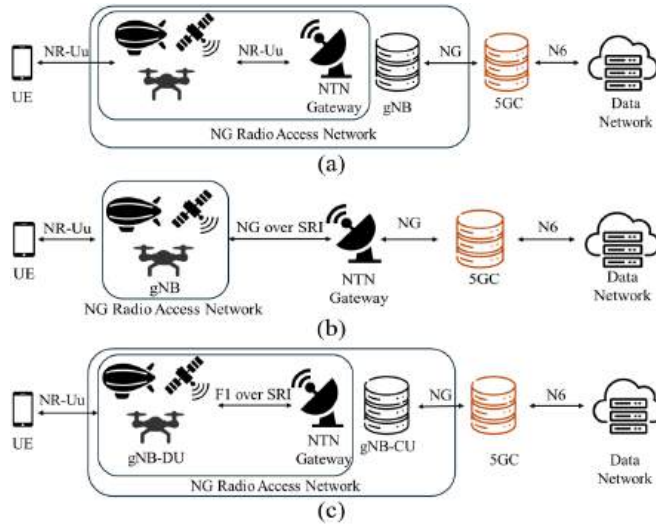


Fig. 2. NTN-based NG-RAN architecture: (a) Transparent NTN-based architecture. (b) Regenerative NTN-based (full gNB) architecture. (c) Regenerative NTN-based (gNB-DU) architecture.

By combining these components, NTN systems can provide connectivity to remote and underserved areas, extending the reach of mobile networks and enabling a wide range of applications. Fig. 2(a) illustrates the transparent-payload-based NTN architecture. The NTN platform acts as a simple relay, forwarding NR signals between the NTN gateway and the UE. This eliminates the need for modifications to the NG-RAN architecture to accommodate satellite access. Both the feeder and service links utilize the NR-Uu interface. The NTN gateway forwards the NR signal received from the NR-Uu interface to the gNB. The gNB connects to the 5G core network (5GC) via the NG interface, while the N6 interface facilitates communication between the 5GC and the data network. Fig. 2(b) depicts an example of the regenerative-payload-based NTN architecture where the NTN platforms act as the gNB. The NR-Uu interface facilitates communication between the UE and the satellite. The NG interface, leveraging a satellite radio interface (SRI), directly connects the NTN platform to the 5GC. Alternatively, shown in Fig. 2(c) is a regenerative-payload-based NTN architecture where the NTN platforms act as a gNB-DU. NTN platforms are connected to ground gNB-CU by a F1 interface over SRI. The NR-Uu interface is used between the UE and the onboard gNB-DU, while the NG interface connects the ground-based gNB-CU to the 5GC.

In addition, 3GPP TR-38.821 [36] explores multi-connectivity, which enables devices simultaneously to access both NTNs and terrestrial networks, or even multiple NTN NG-RANs. Fig. 3(a) illustrates a scenario in which a ground-based UE is connected to the 5GC through both a transparent NTN-based NG-RAN and a ground network. The Xn interface serves as the connection point between the two gNBs, facilitating seamless communication and potential load balancing between the two networks. Fig. 3(b) depicts a scenario where two transparent NTN-based NG-RANs are combined. This hybrid approach can leverage the strengths of different satellite constellations. For instance, a combination of GEO and LEO

satellites can be employed. LEO satellites, with their lower latency, are ideal for supporting delay-sensitive applications, while GEO satellites can provide additional bandwidth to meet high-throughput requirements. By combining these two technologies, a more robust and versatile NTN network can be realized.

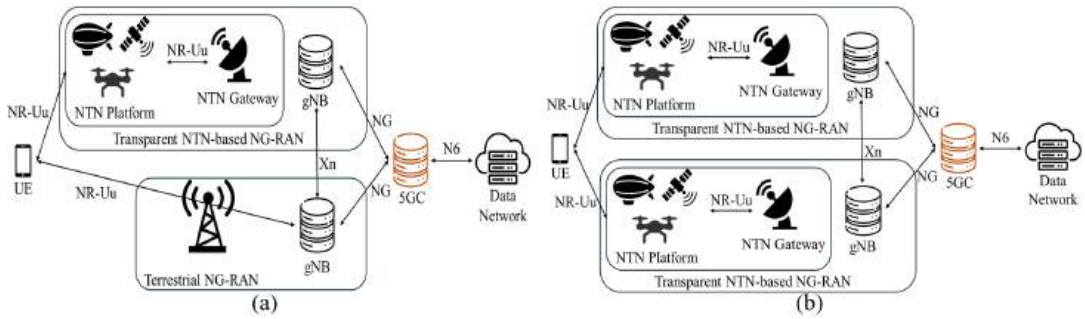


Fig. 3. Multi-connectivity involving transparent NTN-based NG-RANs. (a) Single transparent NTN-based NG-RAN. (b) Two transparent NTN-based NG-RAN.

Multi-connectivity can be implemented between regenerative NTN-based NG-RANs and terrestrial networks. Fig. 4(a) illustrates a scenario where a regenerative NTN-gNB-based NG-RAN is combined with a terrestrial NG-RAN, where the NTN platform assumes all of the functionalities of a gNB. Fig. 4(b) depicts a combination of two regenerative NTN-gNB-based NG-RANs.

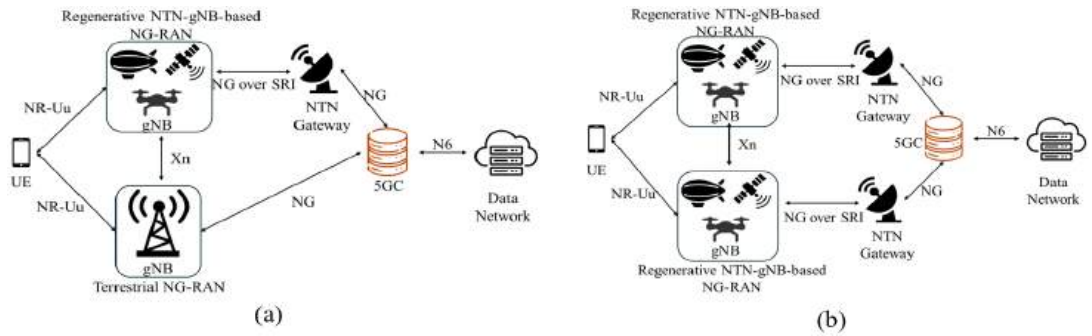


Fig. 4. Multi-connectivity involving regenerative NTN-gNB-based NG-RAN. (a) Single regenerative NTN-gNB-based NG-RAN. (b) Two regenerative NTN-gNB-based NG-RANs.

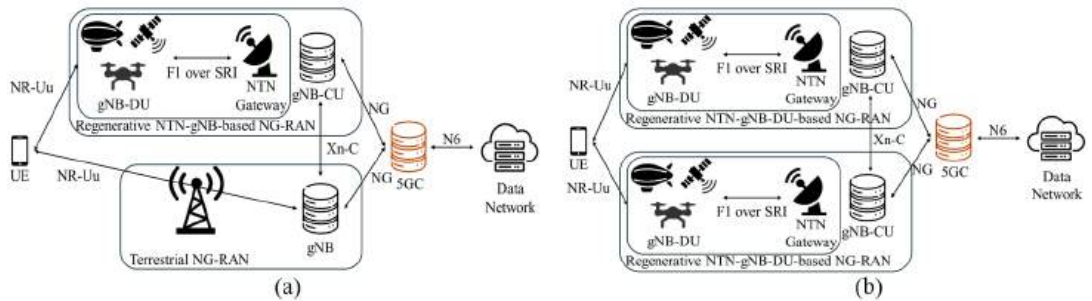


Fig. 5. Multi-connectivity involving regenerative NTN-gNB-DU-based NG-RAN. (a) Single regenerative NTN-gNB-DU-based NG-RAN. (b) Two regenerative NTN-gNB-DU-based NG-RANs.

Finally, Fig. 5(a) illustrates a scenario in which a regenerative NTN-gNB-DU-based NG-RAN is combined with a terrestrial network. This architecture utilizes a functional split, with the NTN platform

acting as a gNB-DU while the corresponding central unit is located on the ground. This approach can be particularly useful for providing services to underserved areas. Fig. 5(b) demonstrates a multi-connectivity scenario involving two regenerative NTN-gNB-DU-based NG RANs.

3.3 Open Issues

3.3.1 Integration with RIS technologies

RISs are planar surfaces composed of many passive elements that can be electronically controlled to redirect or modify electromagnetic waves, enhancing coverage and signal quality levels [37, 38]. By smartly tuning the reflection characteristics, RISs can significantly improve wireless network performance capabilities with minimal energy consumption. RISs have been proposed for inclusion in the 3GPP Release 19 standardization [39]. In NTN communications, multi-hop transmissions are often necessary due to the usage of diverse NTN platforms and gateways. RIS-reflected signals can play a crucial role in optimizing these transmissions. By constructively combining reflected signals at the intended UE or destructively interfering with unintended receivers, RISs can enhance the signal quality and reduce interference. RISs can dynamically adjust reflective elements to create virtual line-of-sight (LoS) links, extending the coverage of NTN communications. Therefore, considering RISs in future 3GPP NTN standardization efforts is essential to unlock the full potential of this technology.

3.3.2 Integration with RSMA technologies

RSMA divides data into multiple layers or streams to optimize spectrum and power utilization as well as interference management [40]. Its flexible structure and interference cancellation mechanisms enable higher reliability and better throughput under varying channel conditions. Due to the high mobility of NTN platforms and the varying distances between them, accurate channel estimation remains a challenging task [41]. Atmospheric conditions, such as rain, fog, and other weather-related factors, further complicate the channel estimation process. RSMA has emerged as a promising solution to address these challenges. By decoding part of the interfering signal and treating the rest as noise, RSMA offers a flexible approach to managing interference and improving NTN communication performance [42, 43]. However, the integration of RSMA into the 3GPP NTN architecture introduces additional complexities, such as optimizing common rate and transmit power allocation operations. Careful consideration of these factors is crucial to maximize the benefits of RSMA in NTN systems.

3.3.3 Grant-free random access

Due to the widespread distribution of UEs within the coverage of NTN platforms, achieving massive connectivity with low latency remains a significant challenge [44]. Traditional grant-based random access in NTNs relies on complex scheduling and resource allocation processes, which can introduce delays and increase the signaling overhead [45, 46]. In contrast, grant-free random access allows devices to transmit data immediately without requiring prior access approval, significantly reducing the latency and enabling faster, more efficient communication [47, 48]. This approach is particularly well suited for NTN systems with extensive coverage areas. To support massive connectivity in NTN systems, the standardization of grant-free random access within NTN architectures could be essential.

3.4 Summary

This section provides a comprehensive analysis of the classification, deployment, and integration of NTNs within 5G frameworks. NTN platforms are classified into spaceborne types (GEO, MEO, and LEO satellites) and airborne types (UAS and HAPs), each defined by distinct coverage areas, altitudes, and application scenarios. The discussion highlights the integration of NTNs with 5G NR architectures, focusing on transparent payload platforms that relay signals and regenerative payload platforms capable of advanced signal processing.

The section also examines the key features of 3GPP standards, such as multi-connectivity, all of which enable seamless interoperability between NTN and terrestrial networks while optimizing resource utilization. Emerging technologies such as RIS and RSMA are explored for their ability to enhance coverage, improve spectrum efficiency, and manage interference in NTN systems. Furthermore, grant-free random access is identified as a promising solution to support massive connectivity and low-latency communication, ensuring the scalability and performance of next-generation NTN deployments.

4. Standards for NTN Communications

As the standards of 3GPP continues to evolve, challenges and considerations for NTN communications have emerged across different releases. Fig. 6 presents a timeline of the 3GPP NTN standardization process, offering a structured overview of the evolution of NTN technologies from Releases 15 to 19. It highlights the contributions of two key technical specification groups within 3GPP—radio access network (RAN) and service and system aspects (SA)—which have played pivotal roles in advancing NTN capabilities [49]. Additionally, Table 3 summarizes the specific NTN issues addressed in each release [50–67].

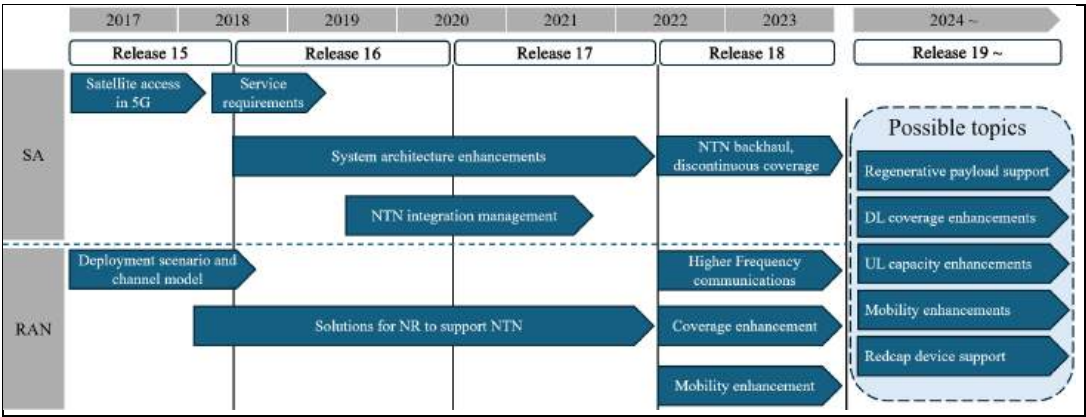


Fig. 6. 3GPP NTN standard timeline.

4.1 The 3GPP NTN Issues by Release

4.1.1 Release 15

TR 38.811 [50] investigates the integration of NTN into 5G NR, focusing on satellite and aerial platforms. It details deployment scenarios, system parameters, and channel models for NTNs, including geostationary and non-geostationary satellites, and HAPs. The study highlights the role of NTNs in enhancing 5G coverage in underserved areas, ensuring service continuity and supporting high-speed mobility and IoT use cases. Architectural options such as transparent and regenerative payloads are explored, along with unique challenges such as Doppler effects and propagation delays. This framework aims to expand 5G capabilities through NTN integration.

4.1.2 Release 16

TR 22.822 [51] examines the integration of satellite access in 5G systems, focusing on enabling service continuity, ubiquity, and scalability. The study outlines key use cases, such as bridging connectivity gaps in remote or underserved regions, supporting IoT applications, enabling seamless roaming between terrestrial and satellite networks, and ensuring reliable communication in disaster scenarios. It emphasizes the role of satellite networks in complementing terrestrial systems through multicast, broadcast capabilities, and extended backhaul services. The report identifies potential requirements and challenges, including latency considerations, quality of service (QoS) adaptations, and regulatory compliance, paving

the way for satellite integration to enhance global 5G network coverage and resilience.

TR 38.821 [36] presents solutions for adapting the 5G NR protocol to support NTN, focusing on satellite access and related architectures. The study explores transparent and regenerative payload models for satellites, covering scenarios with geostationary and non-geostationary orbits. It addresses challenges such as propagation delays, Doppler effects, and mobility management, proposing modifications to physical and higher-layer protocols to ensure reliable communication. By integrating NTN with terrestrial networks, the report aims to extend 5G coverage, enhance service continuity, and enable advanced use cases, including IoT and high-speed mobility.

Table 3. Summary of 3GPP NTN issues by release

	Description
Release 15	Integrates NTN into 5G NR with satellite/aerial platforms and channel models, covering Doppler, propagation delays, and payload options to expand coverage and support IoT/mobility [50].
Release 16	Adds satellite access for continuity and scalability, bridging remote gaps, handling IoT, roaming, QoS, and regulatory needs [51]. Adapts 5G NR for NTN via modified protocols addressing Doppler, delays, and payload models for broader coverage [36].
Release 17	Defines UAS connectivity/authorization under UAV traffic management, ensuring safe UAV operations with robust C2 [52]. Focuses on orchestrating satellite components in 5G, covering network slicing, resource management, and latency [53]. Integrates UAS via SEAL services, handling UAV communication, location, and QoS for reliable operations [54]. Enhances UAV architecture with authentication, geofencing, and C2 flows for secure aviation compliance [55]. Examines PLMN selection for satellite coverage, focusing on cross-border scenarios and GNSS-based locations [56]. Defines UAS application layers, addressing QoS, event reporting, and UAV functionalities in 5G networks [57].
Release 18	Introduces new IoT NTN bands for LTE IoT, detailing frequency ranges, channel arrangements, and coexistence issues [58]. Outlines NTN requirements in NR, emphasizing regulatory location accuracy and large coverage areas [59]. Explores satellite backhaul for remote connectivity, covering delay management, QoS control, and edge computing [60]. Defines the minimum RF specs for satellite-supported E-UTRA, addressing Doppler, interference, and power control schemes [61]. Describes the mIAB architecture for mobile base stations, including handovers, synchronization, and hierarchical protocols [62]. Focuses on RF coexistence for NTN in NR, proposing interference mitigation methods and power-control strategies [63].
Release 19	Enhances UAV features (geofencing, detect-and-avoid), ensuring network-assisted safety in regulated airspace [64]. Covers satellite communication use cases (S&F, positioning), aiding remote IoT and disaster scenarios [65]. Manages regenerative/transparent payloads and frequent satellite handovers, supporting batch pre-configuration [66]. Handles mobile IAB node management, including power-up, mobility, and real-time configuration updates [67].

4.1.3 Release 17

TR 23.754 [52] studies system enablers for supporting UAS connectivity, identification, and tracking within 3GPP networks, addressing UAV controllers and UAV identification, authorization, and authentication by UAV traffic management (UTM). It explores solutions for UAV tracking, including UAV controller associations, and defines mechanisms to handle unauthorized UAVs through revocation procedures. The report highlights key use cases for UAV communication, including user plane-based connectivity for command-and-control (C2) communication, UAV identification, and tracking across various network architectures, enhancing operational safety and compliance in regulated airspaces.

TR 28.808 [53] focuses on the management and orchestration aspects of integrating satellite components in a 5G network. It explores reference architectures, scenarios, and use cases, examining the technical challenges and requirements for satellite integration. Various architectural models are discussed, including 5G networks with combined terrestrial and satellite RANs, and scenarios for satellite-enabled network slices. Use cases address satellite and terrestrial RANs for network slicing, resource management, and delay monitoring, with potential solutions to optimize performance and to support seamless integration across non-geostationary and geostationary orbits. The report identifies potential modifications to 5G management systems, proposing specific enhancements to manage satellite-induced latency and mobility efficiently.

TR 23.755 [54] focuses on integrating UAS into 3GPP networks, addressing UAV communication, location management, and QoS provisioning through service enabler architecture layer (SEAL) services. Key topics include broadcast communications, UAV identification, and C2 communication mode switching to ensure reliable UAV control during operations. Solutions include network-assisted positioning, QoS coordination, and real-time monitoring to enhance the safety and efficiency of UAS operations in regulated airspaces.

TS 23.256 [55] outlines architectural enhancements for UAS to support connectivity, identification, and tracking in compliance with aviation requirements. It introduces mechanisms for UAV authentication and authorization, including UAV-to-ground identification, geofencing, and C2 communication. The document details functional entities and information flows within 5G and LTE systems, emphasizing integration with UAS service suppliers (USS) for effective traffic management and regulatory compliance. It addresses UAV tracking, flight authorization, and the management of unauthorized UAVs, ensuring secure and efficient UAS operations within the 3GPP system.

TR 24.821 [56] focuses on public land mobile network (PLMN) selection for satellite access in 5G networks. It examines key challenges in integrating satellite access within PLMN selection, particularly across international and cross-border areas. Scenarios are explored where satellite cells may span multiple countries or international zones, necessitating adjustments to standard PLMN selection procedures. Key issues include determining UE locations, handling global/shared PLMN IDs, and ensuring PLMN selection in international regions. Solutions proposed involve using GNSS for accurate location determinations, adapting PLMN IDs to recognize international zones, and revising network protocols to accommodate satellite access efficiently. The report also details potential modifications to UE and network systems to support seamless 5G and satellite integration.

TS 23.255 [57] defines the functional architecture and information flows for the UAS application enabler layer within 3GPP systems. It supports UAS connectivity, identification, and tracking through an architecture comprising UAS application-specific and UAS application enabler layers. The document describes QoS provisioning for C2 communication, UAV location deviation monitoring, and UAV event reporting, emphasizing seamless communication between UAVs and integration with UAS traffic management systems. It outlines the role of SEAL services, including location and group management, to support advanced UAS functionalities such as real-time monitoring, C2 communication switching, and group-based QoS management. This framework aims to ensure efficient, secure, and compliant UAS operations within 5G networks.

4.1.4 Release 18

TR 36.764 [58] discusses the introduction of additional IoT NTN bands, particularly the extended L-

band (band B253) and the L+S-band (band B254), to accommodate LTE IoT and NTN operations. Band B253 includes uplink frequencies from 1,668 MHz to 1,675 MHz and downlink frequencies from 1,518 MHz to 1,525 MHz, while band B254 utilizes uplink frequencies from 1,610 MHz to 1,626.5 MHz and downlink frequencies from 2,483.5 MHz to 2,500 MHz. The report highlights flexible channel arrangements and frequency separation considerations to mitigate interference and optimize performance, particularly for NB-IoT and eMTC coexistence with existing services. These frequency extensions aim to enhance NTN capabilities for IoT use cases and support expanded 5G network functionalities.

TR 38.882 [59] examines the requirements and use cases for NTN within NR, focusing on improving location accuracy and reliability for regulatory services. Key use cases include emergency calls, lawful intercepts, public warning systems, and charging/tariff notifications, where network-based solutions are required to verify UE locations alongside GNSS data. The study emphasizes the importance of ensuring accuracy within 5–10 km for NTN cells, which may span large geographic areas, to meet regulatory and operational demands.

TR 23.700-27 [60] explores the integration of satellite backhaul into the 5G system to enhance connectivity, especially in remote and underserved areas. It addresses key challenges such as dynamic delay management, QoS control, and satellite edge computing through on-board user plane functions (UPFs). Proposed solutions include enabling local data switching to reduce latency levels, supporting edge computing services, and optimizing policy control based on satellite backhaul characteristics. These innovations aim to improve 5G network resilience and scalability while enabling advanced use cases such as IoT and edge services in satellite-connected networks.

TS 36.102 [61] outlines the minimum RF characteristics and performance requirements for UE-supporting satellite access in evolved universal terrestrial radio access (E-UTRA) systems. It specifies operational guidelines for various UE categories, including M1, NB1, and NB2, covering aspects such as operating bands, channel arrangements, transmitter and receiver characteristics, and signal quality. The document addresses unique challenges such as Doppler effects, spurious emissions, and adjacent channel interference, ensuring compliance with rigorous standards for reliable satellite communication in 5G environments. It also emphasizes power control, modulation quality, and spectral efficiency to optimize UE performance in satellite-connected networks.

TS 38.401 [62] focuses on the technical architecture and operational principles of mobile integrated access and backhaul (mIAB). mIAB supports mobility in IAB nodes, allowing seamless integration of mobile base stations such as those mounted on vehicles or drones. It highlights critical functionalities such as dynamic handovers, synchronization, backhaul adaptability, and hierarchical architecture to support diverse deployment scenarios. Key protocols and interfaces such as NR-Uu, F1, Xn, and NG are outlined to enable robust connectivity for mobile IAB-DUs and IAB-MTs. The specifications also address operational challenges such as link recovery, resource management, and compliance with latency and quality standards, enabling efficient and scalable mobile network extensions.

TR 38.863 [63] focuses on the RF and coexistence aspects of NTN in NR, addressing the impact of satellite and terrestrial network interactions. It proposes solutions to minimize interference, particularly in adjacent bands, by simulating coexistence scenarios for both satellite and terrestrial operations in the 2GHz and 17/27GHz ranges. The study includes considerations of antenna and beamforming models, propagation effects, and transmission power control to ensure that NTNs and terrestrial networks can operate efficiently without degraded performance outcomes on overlapping or adjacent frequency bands.

4.1.5 Release 19

TR 22.843 [64] examines use cases and requirements for enhancing 5G system support for UAVs, focusing on pre-flight preparation, inflight operations, and safety mechanisms such as geofencing and detect-and-avoid (DAA). It highlights the importance of UAV detection, ensuring that UAVs operate within regulated visual LoS constraints and supporting UAV inflight operations through real-time monitoring and QoS adjustments based on network conditions. The study also explores network-assisted

UAV DAA functionality and multi-network UAV operations, ensuring reliable communication and operational safety across diverse scenarios.

TR 22.865 [65] investigates several satellite communication use cases within the 5G system, focusing on S&F operations, UE-satellite-UE communication, and satellite-aided positioning. Key use cases include enabling delay-tolerant communication services in remote areas, inter-satellite data transfers, and efficient satellite communication without reliance on ground networks. The study highlights how S&F mechanisms can support IoT devices, such as animal tracking and disaster monitoring, while positioning enhancements allow for improved location accuracy in GNSS-denied environments. These use cases emphasize the need for enhanced satellite access capabilities within the 5G system to meet diverse global communication needs.

TR 28.874 [66] investigates the management aspects of NTN in 5G, focusing on managing the interactions between satellite and ground systems for regenerative and transparent payloads. Key use cases include managing connections and associations between satellite and core network nodes, handling S&F satellite operations, and supporting UE-satellite-UE communication. The study emphasizes solutions such as batch pre-configurations of network associations to address frequent connection changes due to satellite movement and optimized management schemes for dynamic mobility coordination.

TR 28.875 [67] focuses on managing mobile IAB nodes in 5G networks, addressing use cases such as power-up, mobility, and reconfiguration. It outlines requirements for secure, location-based IP configurations and connectivity to operations, administration, and maintenance (OAM) systems, as well as solutions for dynamic configuration updates during mobility. Proposed methods include configuration mapping tied to location, PDU session-based IP acquisition, and notification mechanisms for real-time updates.

4.2 Performance Requirements for NTN Communications

4.2.1 Airborne communication requirements

TS 22.125 [68] demonstrates the performance requirements of airborne communications. Table 4 outlines the performance requirements for various airborne services, focusing on uplink and downlink data rates, along with latency constraints. High-bandwidth services such as 8K video live broadcast and 4x4K AI surveillance demand significant uplink data rates of 100 Mb/s and 120 Mb/s, respectively, with 200 ms and 20 ms latency limits. Surveillance services also require a substantial downlink capacity of 50 Mb/s for real-time AI-based monitoring. Remote UAV control with HD video requires a 25 Mb/s uplink with 100 ms latency and 300 kb/s downlink with 20 ms latency. In contrast, real-time video and periodic still photos require an uplink data rate of 0.06 Mb/s and 1 Mb/s with 100 ms and 1 second latency. Video streaming applications depend on uplink rates of 4 Mb/s for 720p and 9 Mb/s for 1080p resolutions, with latency of 200 ms.

Table 4. Performance requirements for various airborne services

Service	Uplink		Downlink	
	Data rate	Latency	Data rate	Latency
8K video live broadcast	100 Mb/s	200 ms	600 kb/s	20 ms
4x4K AI surveillance	120 Mb/s	20 ms	50 Mb/s	20 ms
Remote UAV controller through HD video	25 Mb/s	100 ms	300 kb/s	20 ms
Real-time video	0.06 Mb/s	100 ms	-	-
Video streaming	4Mb/s→720p	200 ms	-	-
	9 Mb/s→ 1080p			
Periodic still photos	1 Mb/s	1 s	-	-

4.2.2 Spaceborne communication requirements

TS 22.261 [69] demonstrates the performance requirements of spaceborne communications. Table 5 provides performance requirements for various spaceborne communication scenarios, detailing downlink and uplink data rates, UE speeds, and UE types. Pedestrian scenarios require a 1 Mb/s downlink and a 100 kb/s uplink for handheld devices, with UEs moving at walking speeds. In public safety scenarios, slightly higher rates of 3.5 Mb/s for both uplink and downlink are needed. For vehicular connectivity, where UEs are vehicle-mounted and travel at speeds of up to 250 km/h, the downlink and uplink data rates rise to 50 Mb/s and 25 Mb/s, respectively. Airplane connectivity represents the highest performance requirements, with a need for 360 Mb/s downlink and 180 Mb/s uplink rates per plane, supporting speeds of up to 1,000 km/h for airplane-mounted UEs. Other use cases include stationary UEs, such as building-mounted devices, requiring 50 Mb/s downlink and 25 Mb/s uplink rates. Video surveillance scenarios demand 0.5 Mb/s downlink and a 3 Mb/s uplink rates, serving stationary and moving UEs at speeds up to 120 km/h. Finally, narrowband IoT connectivity is tailored for low-power IoT devices with minimal data requirements, needing just a 2 kb/s downlink and a 10 kb/s uplink for devices operating at speeds of up to 100 km/h.

Table 5. Performance requirements for spaceborne scenarios

Service	Downlink data rate	Uplink data rate	UE speed	UE type
Pedestrian	1 Mb/s	100 kb/s	Pedestrian	Handheld
Public safety	3.5 Mb/s	3.5 Mb/s	100 km/h	Handheld
Vehicular connectivity	50 Mb/s	25 Mb/s	Up to 250 km/h	Vehicle mounted
Airplane connectivity	360 Mb/s/plane	180 Mb/s/plane	Up to 1,000 km/h	Airplane mounted
Stationary	50 Mb/s	25 Mb/s	Stationary	Building-mounted
Video surveillance	0.5 Mb/s	3 Mb/s	Up to 120 km/h or stationary	Vehicle-mounted or fixed installation
Narrowband IoT connectivity	2 kb/s	10 kb/s	Up to 100 km/h	IoT

4.3 Potential Focus Areas for Future 3GPP NTN Specifications

4.3.1 Regenerative payload support

Regenerative payloads represent a promising avenue for future NTN specifications. By enabling packet switching within the payload itself, these systems can bypass terrestrial networks, potentially leading to significant latency reductions and improved network efficiency [49]. While previous 3GPP releases have primarily focused on transparent payloads, the incorporation of regenerative payload support in future releases could revolutionize NTN network architectures and capabilities.

4.3.2 Uplink/downlink enhancements

As the number of UEs connected to NTN platforms increases, there is a growing need for advancements in both uplink and downlink communication. Release 18 marked a significant step forward in expanding uplink coverage, but further improvements are necessary to address potential bottlenecks and capacity limitations. Future releases should explore techniques such as advanced modulation and coding schemes, dynamic resource allocation, and interference mitigation strategies to enhance both uplink and downlink performance.

4.3.3 Mobility enhancements

Efficient handover between NTN platforms is crucial for seamless user experience outcomes and network performance guarantees. However, the complexity of this process, particularly in terms of interference management and time synchronization, poses significant challenges. Future specifications should prioritize the development of advanced handover techniques that minimize disruptions, reduce latency, and ensure smooth transitions between different NTN platforms.

4.3.4 RedCap support

The integration of RedCap devices into NTN networks presents unique opportunities and challenges. While RedCap devices offer significant advantages in terms of power consumption and cost-effectiveness, their deployment in NTN environments requires careful consideration of factors such as latency, Doppler shift, and signaling overhead. Future specifications should explore the proper adaptations necessary to enable seamless and reliable communication between RedCap devices and NTN platforms, thereby expanding the range of IoT applications that can benefit from NTN technology.

4.4 Summary

This section outlines the evolution of NTN integration within the 3GPP framework across Releases 15–19. Release 15 established the foundational concepts for NTN integration with 5G NR, including deployment scenarios, system parameters, and channel models. Release 16 built upon these foundations by addressing satellite-enabled backhaul, IoT applications, and regulatory considerations, while ensuring service continuity and QoS adaptations. Release 17 introduced advancements in UAV connectivity, focusing on identification, tracking, and C2 mechanisms, along with architectural refinements to support terrestrial satellite coexistence. Release 18 expanded NTN functionalities with the addition of new frequency bands, advanced satellite backhaul solutions, and enhanced IoT support, prioritizing latency reduction and satellite edge computing. Finally, Release 19 further developed UAV use cases with geofencing, DAA features, and inter-satellite communication, while improving regenerative and transparent payload management. Together, these advancements significantly enhance the integration of NTN systems with terrestrial networks, fostering a robust and globally interconnected 5G and beyond 5G ecosystem.

5. Service Scenarios in NTN Communications

This section explores the service scenarios outlined in technical specifications TS 22.843 [64] and TS 22.865 [65]. We also discuss open issues related to these scenarios.

5.1 Airborne Communications Scenarios

Fig. 7 illustrates two key scenarios for airborne communications: supporting UAV pre-flight preparation and network-assisted UAV DAA [64].

Many industries utilize UAVs along predefined routes. To ensure reliable and stable operations for multiple UAVs, especially when routes overlap, UAV operators require access to network information from 3GPP networks and UTM systems. This enables proactive flight path adjustments [70], ensuring real-time performance, stability, and uninterrupted operation in the flight control link. As depicted in Fig. 7(a), a service scenario for supporting UAVs pre-flight involves the following steps [64]. First, UAV A plans a flight from point X to Y, while UAV B plans to fly from point X to Z. Second, UAV A and UAV B provide the UTM system with the planned flight route information, including waypoints and the required QoS. Third, the UTM system, in conjunction with the 5G system, reviews the necessary QoS information and predictions/statistics, such as the end-to-end latency and quality of experience (QoE) data. Fourth, the 5G system is at this point able to verify whether the required QoS is available for UAV A's designated flight path. If not, it can reconfigure resources to provide QoS, while UAV B's path is optimized based on the available QoS. Fifth, the UTM system determines the final flight routes and transmits the routes to UAV A and UAV B. Finally, UAV A and UAV B receive the flight route information from the UTM system via the 5G system and commence their missions.

In addition, a network-assisted approach to UAV DAA is crucial for enhancing operational safety by facilitating effective DAA measures [71]. This scenario involves both 5G-registered and unregistered

UAVs operating simultaneously. Modern DAA systems employ UAV sensing technology to detect unregistered UAVs and subsequently send DAA notifications to improve situational awareness and safety. As shown in Fig. 7(b), the 3GPP standard outlines a use case where networks incorporate ground-based sensing to enhance UAV DAA capabilities further. This ground-assisted layer proactively improves flight safety by preventing collisions. The service scenario flow is as follows. First, UAV A registers with the 5G network and communicates with the UTM system through the 5G network, while UAV B establishes direct communication with the UTM system. Second, the UTM system gathers information about UAV A and UAV B within a specific area via base stations and uses it for DAA operations. Third, the UTM system updates its records. Then, it sends DAA notifications to UAV A and UAV B operating in the area. Eventually, UAV A and UAV B receive the notifications and respond with appropriate actions.

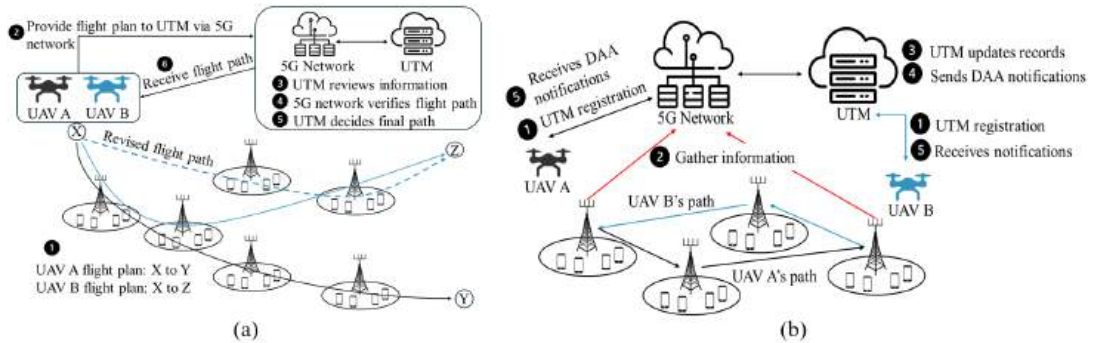


Fig. 7. Airborne platform service scenarios. (a) Supporting UAV pre-flight preparation. (b) Network-assisted UAV DAA.

These scenarios have significant real-world implications across various industries. For instance, UAV-based delivery systems leverage pre-flight planning services to optimize routes, reduce travel times, and avoid congestion. In industrial zones and large-scale construction sites, network-assisted DAA technologies are essential for ensuring safety during UAV deployments for inspection, mapping, and delivery tasks. Furthermore, in disaster-response scenarios, UAVs can navigate efficiently, avoid collisions, and deliver critical supplies to inaccessible areas, highlighting their pivotal role in emergency operations. In these applications, UAV operators share flight routes and relevant service requirements with the 5G and UTM systems during the pre-flight stage. This process ensures stable network performance, real-time updates, and dynamic resource allocation tailored to each UAV's operational needs. Additionally, the network facilitates detection and avoidance mechanisms by monitoring both registered and unregistered UAVs, enhancing situational awareness and safety. By integrating these capabilities, UAV operations can feasibly expand into urban areas, supporting applications such as package delivery, public safety, and smart city infrastructure management.

However, the increasing use of UAVs presents several challenges and limitations. Managing overlapping flight paths in densely populated areas and ensuring stable control links under high network demand remain critical challenges. A significant limitation is the reliance on extensive network infrastructure, particularly in rural or remote areas where 5G coverage may be sparse or unreliable. Additionally, UAV DAA systems need to evolve with advanced sensing technologies and AI-driven decision-making to handle large-scale deployments effectively. Latency in updating and distributing flight route information among multiple UAVs in high-density areas poses another limitation, potentially impacting real-time operations and mission-critical tasks. Addressing these challenges will require innovative approaches in network design and resource management as well as the seamless integration of NTN with terrestrial systems.

5.2 Spaceborne Communications Scenarios

Fig. 8 illustrates two key spaceborne communication scenarios: UE-satellite-UE phone calls [65] and service continuity for urban air mobility (UAM) communication [66].

Traditional wireless networks often struggle to provide coverage in rural or disaster-affected areas due to infrastructure limitations or damage. Satellites can extend network coverage, enabling UEs in these regions to communicate effectively [72]. In disaster situations, satellite connectivity can facilitate rescue operations by connecting to the UE closest to the affected area. Fig. 8(a) depicts a service scenario where satellites enhance connectivity to locate and rescue UEs in disaster-affected areas. Initially, UE A, dealing with injuries in an unserved area, initiates an emergency call to a rescue center for medical assistance. Then, using location data from the satellite networks, the rescue center determines that UE B is the nearest available responder and directs the call to UE A. In the end, UE B receives the call through satellite connectivity and begins tracking UE A's position using real-time location updates. To reduce communication delays, the call is routed directly via a single satellite, bypassing terrestrial networks.

Given that UAM enables the transportation of passengers and cargo, reliable communication is essential for UAM safety and control [73]. UAM vehicles must access real-time information about other airborne vehicles, weather conditions, and operational zones for effective management and navigation. Moreover, these vehicles must maintain stable communication even beyond the coverage of ground base stations to ensure uninterrupted operation. Fig. 8(b) illustrates a service scenario where two UAM vehicles exchange data and communicate seamlessly. At the beginning, UAM A operates within the satellite network coverage range, while UAM B is initially connected to the ground network. UAM B receives flight information about UAM A through both satellite and terrestrial networks before departure. Next, when UAM B moves to satellite coverage, it switches to satellite communication to receive real-time updates on UAM A. Finally, if UAM B re-enters the terrestrial network's coverage, it exchanges flight information with UAM A via both satellite and terrestrial networks.

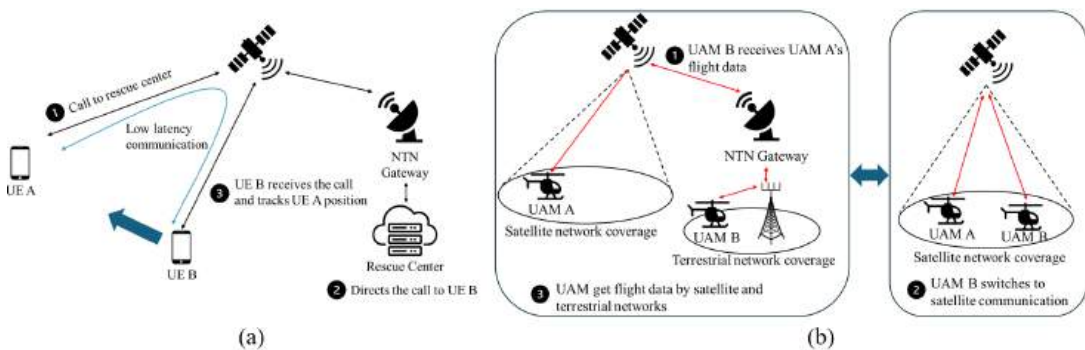


Fig. 8. Spaceborne platform service scenarios: (a) UE-satellite-UE phone call. (b) Service continuity for UAM-to-UAM communication.

The scenarios described above are particularly impactful in disaster management, where rapid satellite-enabled communication can play a critical role in saving lives. NTN technologies facilitate real-time connectivity between rescue teams and affected individuals, ensuring seamless coordination during emergencies. For instance, UAM services can utilize NTN technologies for safe and efficient operations in areas without ground-based network coverage. UAM taxis can be deployed to evacuate individuals from disaster-stricken zones, transporting them to safety while maintaining continuous communication with emergency response teams. Additionally, industries such as the mining and maritime sectors benefit significantly from satellite connectivity. NTN technologies enable real-time communication for remote mining operations and vessels navigating open seas, enhancing both operational efficiency and safety by providing reliable links in otherwise isolated environments.

However, scalability poses significant challenges as the reliance on satellite communication grows. During emergencies in densely populated regions, increased demand could lead to congestion, higher latency, and potential service disruptions. Additionally, ensuring seamless service continuity during transitions between satellite, airborne, and terrestrial networks remains a critical technical limitation, particularly when managing heterogeneous network architectures. These challenges highlight the need for advanced resource management strategies, robust network coordination, and optimization techniques to support large-scale deployments while maintaining reliable and efficient communication.

5.3 Open Issues

5.3.1 NTN-based O-RAN

O-RAN establishes open interfaces and separates hardware and software components within the RAN, enabling greater flexibility and interoperability across vendors. This architecture promotes innovation, reduces costs, and supports scalable deployments, making it well suited to meet the evolving demands of 5G and 6G networks. To enhance O-RAN performance capabilities, the placement of O-CU and O-DU components is crucial in O-RANs [52]. Here, NTNs can be leveraged to enhance the flexibility, scalability, and coverage of O-RAN systems. O-CUs can be implemented on terrestrial gateways or central hubs to handle control plane functionalities, ensuring stable and efficient coordination. O-DUs, on the other hand, can be deployed on NTN platforms such as satellites or UAVs, bringing distributed processing capabilities closer to the edge. This reduces latency and improves real-time data processing. By strategically deploying O-CU and O-DU components on NTN platforms, we can optimize the utilization of communication resources while ensuring compatibility with O-RAN standards.

5.3.2 NTN-based AI-RAN

AI-RANs will be pivotal in 6G networks, enabling smarter decision-making and real-time optimizations [74]. To enhance the real-time adaptability of AI-RANs, NTN platforms can support massive data collection and processing. Latency-sensitive tasks and data collection can be handled locally on NTN platforms, while terrestrial systems manage large-scale analytics and AI model training. This approach ensures responsiveness and scalability, significantly enhancing the importance of NTN assistance as a 3GPP service scenario.

5.3.3 Starlink

Starlink is a prominent LEO satellite constellation designed to deliver high-speed internet access globally [75]. Utilizing thousands of satellites, it aims to provide low-latency broadband connectivity, effectively addressing coverage gaps in regions where terrestrial infrastructure is either unavailable or prohibitively expensive to deploy. This approach enables rapid deployment and near-real-time data transmission, making it particularly valuable for remote and underserved areas.

However, integrating Starlink into existing 5G/6G infrastructures presents significant challenges. Large-scale deployments raise critical questions about inter-satellite link management and global spectrum coordination. Seamless handovers between satellites and terrestrial networks require advanced mobility management techniques to ensure uninterrupted service. Additionally, achieving interoperability with 3GPP NTN standards and O-RAN architectures adds complexity to the integration process. As the constellation expands, further challenges include optimizing traffic routing, mitigating interference across multiple LEO systems, and developing robust strategies for satellite collision avoidance. Overcoming these hurdles will be essential for Starlink to fulfill its potential as a high-capacity backbone for next-generation communication networks.

5.3.4 NTN switching

NTN switching refers to the seamless transfer of user devices between different NTN platforms or from an NTN to a terrestrial network [76]. This process is essential for maintaining uninterrupted connectivity,

particularly in scenarios where network footprints shift dynamically, or link conditions vary.

Recognizing NTN switching as a dedicated 3GPP service scenario is crucial for addressing high mobility demand levels and ensuring consistent, high-quality coverage. Effective implementation requires advanced mobility management, efficient resource allocation, and standardized signaling protocols to manage rapid transitions, minimize disruptions, and maintain global connectivity. As 3GPP continues its standardization efforts, developing robust NTN switching capabilities remains a key challenge to achieving seamless and reliable connectivity in next-generation communication networks.

5.4 Summary

This section explores the potential applications of NTNs in airborne and spaceborne communication scenarios, emphasizing their advantages and challenges. In airborne communications, key scenarios include UAV pre-flight preparation and network-assisted DAA systems. Pre-flight planning utilizes the 5G system and UTM coordination to optimize flight paths and ensure QoS, while DAA systems enhance safety by detecting unregistered UAVs and facilitating collision avoidance through network-assisted alerts. For spaceborne communications, scenarios such as UE-satellite-UE connectivity and service continuity for UAM are highlighted. Satellites enable reliable communication in rural and disaster-affected areas, supporting emergency calls and real-time location tracking. In UAM scenarios, satellite networks ensure uninterrupted operations by enabling real-time data exchanges and seamless transitions between terrestrial and satellite networks.

The section also discusses open issues, including integrating NTN platforms into O-RAN and AI-RAN architectures to enhance flexibility and enable real-time optimization. Additional challenges include leveraging NTN-based AI-driven decision making, addressing Starlink's role in providing global coverage, managing NTN switching, and achieving efficient spectrum coordination. Overcoming these challenges will be critical to realizing seamless, scalable, and efficient NTN services in future communication networks.

6. Conclusion

NTNs represent a transformative innovation in the global communication infrastructure, addressing the limitations of terrestrial systems by enabling connectivity in remote, underserved, and disaster-affected areas. These networks facilitate critical applications such as the IoT, autonomous systems, emergency services, and global mobility, underscoring their vital role in bridging the digital divide and supporting next-generation connectivity demands. This paper has provided a comprehensive review of NTNs, focusing on their architectural advancements, the progressive evolution of 3GPP standards from Release 15 to Release 19, and their integration with terrestrial networks. The standardization efforts highlight the importance of unified frameworks for ensuring seamless interoperability and enabling diverse applications. Key scenarios discussed include UAV operations, maritime connectivity, and environmental monitoring, demonstrating the capacity of NTNs to address real-world challenges and foster technological innovation.

While NTNs hold immense promise, significant challenges remain to be addressed. Persistent issues such as latency reduction, energy efficiency, advanced mobility management, and optimized spectrum allocation require focused research to unlock the full potential of these networks. Future directions must also explore the integration of emerging technologies such as RIS, RSMA, and AI to enhance NTN capabilities further. By tackling these challenges and leveraging their unique advantages, NTNs are poised to become a cornerstone of future communication systems, enabling ubiquitous, sustainable, and intelligent global connectivity.

Author's Contributions

Conceptualization, HJP, T-HN, LHP; Funding acquisition, WSC, KMS, LHP; Investigation, HJP, KHS, SYS, YRL, T-HN; Project administration, LHP; Supervision, LHP; Writing of the original draft, HJP, T-HN; Writing of the review and editing, HJP, T-HN, KHS, SYS, YRL, WSC, KMS, LHP.

Funding

This study was supported by the Research Program funded by the Seoul National University of Science and Technology.

Competing Interests

The authors declare that they have no competing interests.

References

- [1] A. Iqbal, M. L. Tham, Y. J. Wong, A. A. Al-Habashna, G. Wainer, Y. X. Zhu, and T. Dagiuklas, "Empowering non-terrestrial networks with artificial intelligence: a survey," *IEEE Access*, vol. 11, pp. 100986-101006, 2023. <https://doi.org/10.1109/ACCESS.2023.3314732>
- [2] T. H. Nguyen, T. P. Truong, A. T. Tran, N. N. Dao, L. Park, and S. Cho, "Intelligent heterogeneous aerial edge computing for advanced 5G access," *IEEE Transactions on Network Science and Engineering*, vol. 11, no. 4, pp. 3398-3411, 2024. <https://doi.org/10.1109/TNSE.2024.3371434>
- [3] C. K. Wen, L. S. Tsai, A. Shojaeifard, P. K. Liao, K. K. Wong, and C. B. Chae, "Shaping a smarter electromagnetic landscape: IAB, NCR, and RIS in 5G standard and future 6G," *IEEE Communications Standards Magazine*, vol. 8, no. 1, pp. 72-78, 2024. <https://doi.org/10.1109/MCOMSTD.0008.2300036>
- [4] K. Rasilainen, T. D. Phan, M. Berg, A. Parssinen, and P. J. Soh, "Hardware aspects of sub-THz antennas and reconfigurable intelligent surfaces for 6G communications," *IEEE Journal on Selected Areas in Communications*, vol. 41, no. 8, pp. 2530-2546, 2023. <https://doi.org/10.1109/JSAC.2023.3288250>
- [5] C. De Alwis, P. Porambage, K. Dev, T. R. Gadekallu, and M. Liyanage, "A survey on network slicing security: attacks, challenges, solutions and research directions," *IEEE Communications Surveys & Tutorials*, vol. 26, no. 1, pp. 534-570, 2024. <https://doi.org/10.1109/COMST.2023.3312349>
- [6] F. Matera, M. Settembre, A. Rago, A. Fascista, G. Piro, L. A. Grieco, et al., "Terrestrial and non-terrestrial networks for integrated sensing and communication," in *Proceedings of 2024 IEEE International Symposium on Systems Engineering (ISSE)*, Perugia, Italy, 2024, pp. 1-5. <https://doi.org/10.1109/ISSE63315.2024.10741118>
- [7] O. A. Amodu, R. Nordin, N. F. Abdullah, S. A. Busari, I. Otung, M. Ali, and M. Behjati, "Technical advancements towards RIS-assisted NTN-based THz communication for 6G and beyond," *IEEE Access*, vol. 12, pp. 183153-183181, 2024. <https://doi.org/10.1109/ACCESS.2024.3511494>
- [8] M. El Jaafari, N. Chuberre, S. Anjuere, and L. Combelles, "Introduction to the 3GPP-defined NTN standard: a comprehensive view on the 3GPP work on NTN," *International Journal of Satellite Communications and Networking*, vol. 41, no. 3, pp. 220-238, 2023. <https://doi.org/10.1002/sat.1471>
- [9] N. N. Dao, N. H. Tu, T. D. Hoang, T. H. Nguyen, L. V. Nguyen, K. Lee, L. Park, W. Na, and S. Cho, "A review on new technologies in 3GPP standards for 5G access and beyond," *Computer Networks*, vol. 245, article no. 110370, 2024. <https://doi.org/10.1016/j.comnet.2024.110370>
- [10] H. W. Lee, A. Medles, C. C. Chen, and H. Y. Wei, "Feasibility and opportunities of terrestrial network and non-terrestrial network spectrum sharing," *IEEE Wireless Communications*, vol. 30, no. 6, pp. 36-42, 2023. <https://doi.org/10.1109/MWC.001.2300209>
- [11] F. Rinaldi, H. L. Maattanen, J. Torsner, S. Pizzi, S. Andreev, A. Iera, Y. Koucheryavy, and G. Araniti, "Non-terrestrial networks in 5G & beyond: a survey," *IEEE Access*, vol. 8, pp. 165178-165200, 2020. <https://doi.org/10.1109/ACCESS.2020.3022981>
- [12] M. M. Qazzaz, S. A. Zaidi, D. C. McLernon, A. M. Hayajneh, A. Salama, and S. A. Aldalameh, "Non-terrestrial UAV clients for beyond 5G networks: a comprehensive survey," *Ad Hoc Networks*, vol. 157, article no. 103440, 2024. <https://doi.org/10.1016/j.adhoc.2024.103440>

- [13] A. Guidotti, S. Cioni, G. Colavolpe, M. Conti, T. Foggi, A. Mengali, G. Montorsi, A. Piemontese, and A. Vanelli-Coralli, "Architectures, standardisation, and procedures for 5G satellite communications: a survey," *Computer Networks*, vol. 183, article no. 107588, 2020. <https://doi.org/10.1016/j.comnet.2020.107588>
- [14] M. M. Saad, M. A. Tariq, M. T. R. Khan, and D. Kim, "Non-terrestrial networks: an overview of 3GPP release 17 & 18," *IEEE Internet of Things magazine*, vol. 7, no. 1, pp. 20-26, 2024. <https://doi.org/10.1109/IOTM.001.2300154>
- [15] X. Lin, S. Rommer, S. Euler, E. A. Yavuz, and R. S. Karlsson, "5G from space: an overview of 3GPP non-terrestrial networks," *IEEE Communications Standards Magazine*, vol. 5, no. 4, pp. 147-153, 2021. <https://doi.org/10.1109/MCOMSTD.011.2100038>
- [16] I. A. Elhaby, Y. Fahmy, and M. Kafafy, "A survey on UAV placement optimization for UAV-assisted communication in 5G and beyond networks," *Physical Communication*, vol. 51, article no. 101564, 2022. <https://doi.org/10.1016/j.phycom.2021.101564>
- [17] G. Geraci, A. Garcia-Rodriguez, M. M. Azari, A. Lozano, M. Mezzavilla, S. Chatzinotas, Y. Chen, S. Rangan, and M. Di Renzo, "What will the future of UAV cellular communications be? A flight from 5G to 6G," *IEEE Communications Surveys & Tutorials*, vol. 24, no. 3, pp. 1304-1335, 2022. <https://doi.org/10.1109/COMST.2022.3171135>
- [18] J. Lee and V. Friderikos, "Interference-aware path planning optimization for multiple UAVs in beyond 5G networks," *Journal of Communications and Networks*, vol. 24, no. 2, pp. 125-138, 2022. <https://doi.org/10.23919/JCN.2022.000006>
- [19] G. Colajanni, P. Daniele, A. Nagurney, L. S. Nagurney, and D. Sciacca, "A three-stage stochastic optimization model integrating 5G technology and UAVs for disaster management," *Journal of Global Optimization*, vol. 86, no. 3, pp. 741-780, 2023. <https://doi.org/10.1007/s10898-023-01274-z>
- [20] H. Zhang, Z. Qi, J. Li, A. Aronsson, J. Bosch, and H. H. Olsson, "5G network on wings: a deep reinforcement learning approach to the UAV-based integrated access and backhaul," *IEEE Transactions on Machine Learning in Communications and Networking*, vol. 2, pp. 1109-1126, 2024. <https://doi.org/10.1109/TMLCN.2024.3442771>
- [21] Y. Shibata, W. Takabatake, K. Hoshino, A. Nagate, and T. Ohtsuki, "Two-step dynamic cell optimization algorithm for HAPS mobile communications," *IEEE Access*, vol. 10, pp. 68085-68098, 2022. <https://doi.org/10.1109/ACCESS.2022.3186003>
- [22] O. Abbasi, A. Yadav, H. Yanikomeroglu, N. D. Dao, G. Senarath, and P. Zhu, "HAPS for 6G networks: potential use cases, open challenges, and possible solutions," *IEEE Wireless Communications*, vol. 31, no. 3, pp. 324-331, 2024. <https://doi.org/10.1109/MWC.012.2200365>
- [23] R. Deka, V. Mishra, I. Ahmed, S. Anees, and M. S. Alam, "On the performance and optimization of HAPS assisted dual-hop hybrid RF/FSO system," *IEEE Access*, vol. 10, pp. 80976-80988, 2022. <https://doi.org/10.1109/ACCESS.2022.3195930>
- [24] R. Alghamdi, H. Dahrouj, T. Y. Al-Naffouri, and M. S. Alouini, "Equitable 6G access service via cloud-enabled HAPS for optimizing hybrid air-ground networks," *IEEE Transactions on Communications*, vol. 72, no. 5, pp. 2959-2973, 2024. <https://doi.org/10.1109/TCOMM.2023.3348842>
- [25] W. Jiang, H. Han, M. He, and W. Gu, "When game theory meets satellite communication networks: a survey," *Computer Communications*, vol. 217, pp. 208-229, 2024. <https://doi.org/10.1016/j.comcom.2024.02.005>
- [26] W. Jiang, "Software defined satellite networks: a survey," *Digital Communications and Networks*, vol. 9, no. 6, pp. 1243-1264, 2023. <https://doi.org/10.1016/j.dcan.2023.01.016>
- [27] C. Ding, J. B. Wang, M. Cheng, M. Lin, and J. Cheng, "Dynamic transmission and computation resource optimization for dense LEO satellite assisted mobile-edge computing," *IEEE Transactions on Communications*, vol. 71, no. 5, pp. 3087-3102, 2023. <https://doi.org/10.1109/TCOMM.2023.3253721>
- [28] T. Q. Duong, L. D. Nguyen, T. T. Bui, K. D. Pham, and G. K. Karagiannidis, "Machine learning-aided real-time optimized multibeam for 6G integrated satellite-terrestrial networks: global coverage for mobile services," *IEEE Network*, vol. 37, no. 2, pp. 86-93, 2023. <https://doi.org/10.1109/MNET.003.2200275>
- [29] G. Cui, P. Duan, L. Xu, and W. Wang, "Latency optimization for hybrid GEO-LEO satellite-assisted IoT networks," *IEEE Internet of Things Journal*, vol. 10, no. 7, pp. 6286-6297, 2023. <https://doi.org/10.1109/JIOT.2022.3222831>
- [30] A. M. Voicu, A. Bhattacharya, and M. Petrova, "Handover strategies for emerging LEO, MEO, and HEO satellite networks," *IEEE Access*, vol. 12, pp. 31523-31537, 2024. <https://doi.org/10.1109/ACCESS.2024.3368503>

- [31] C. T. Nguyen, Y. M. Saputra, N. Van Huynh, T. N. Nguyen, D. T. Hoang, D. N. Nguyen, et al., "Emerging technologies for 6G non-terrestrial-networks: from academia to industrial applications," *IEEE Open Journal of the Communications Society*, vol. 5, pp. 3852-3885, 2024. <https://doi.org/10.1109/OJCOMS.2024.3418574>
- [32] M. M. Azari, S. Solanki, S. Chatzinotas, O. Kodheli, H. Sallouha, A. Colpaert, et al., "Evolution of non-terrestrial networks from 5G to 6G: a survey," *IEEE Communications Surveys & Tutorials*, vol. 24, no. 4, pp. 2633-2672, 2022. <https://doi.org/10.1109/COMST.2022.3199901>
- [33] A. A. Alsaedy and E. K. Chong, "A survey of mobility management in non-terrestrial 5G networks: power constraints and signaling cost," *IEEE Access*, vol. 12, pp. 107529-107551, 2024. <https://doi.org/10.1109/ACCESS.2024.3438613>
- [34] O. B. Yahia, Z. Garroussi, O. Belanger, B. Sanso, J. F. Frigon, S. Martel, A. Lesage-Landry, and G. K. Kurt, "Evolution of high-throughput satellite systems: a vision of programmable regenerative payload," *IEEE Communications Surveys & Tutorials*, vol. 27, no. 3, pp. 1565-1598, 2025. <https://doi.org/10.1109/COMST.2024.3450292>
- [35] T. Naous, M. Itani, M. Awad, and S. Sharafeddine, "Reinforcement learning in the sky: a survey on enabling intelligence in NTN-based communications," *IEEE Access*, vol. 11, pp. 19941-19968, 2023. <https://doi.org/10.1109/ACCESS.2023.3236801>
- [36] 3GPP, "Solutions for NR to support non-terrestrial networks (NTN) (Release 16)," Technical Report #38.821, 2019 [Online]. Available from: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3525>.
- [37] R. Wang, Y. Yang, B. Makki, and A. Shamim, "A wideband reconfigurable intelligent surface for 5G millimeter-wave applications," *IEEE Transactions on Antennas and Propagation*, vol. 72, no. 3, pp. 2399-2410, 2024. <https://doi.org/10.1109/TAP.2024.3352828>
- [38] H. Xie, D. Li, and B. Gu, "Exploring hybrid active-passive RIS-aided MEC systems: from the mode-switching perspective," *IEEE Transactions on Wireless Communications*, vol. 23, no. 9, pp. 11291-11308, 2024. <https://doi.org/10.1109/TWC.2024.3380841>
- [39] N. Li, J. Zhu, P. Li, B. Wang, X. She, and P. Chen, "Considerations on potential standardization work for reconfigurable intelligent surface," in *Proceedings of 2022 IEEE 8th International Conference on Computer and Communications (ICCC)*, Chengdu, China, 2022, pp. 320-324. <https://doi.org/10.1109/ICCC56324.2022.10066051>
- [40] A. S. Shah, M. A. Karabulut, and K. Rabie, "Multiple access schemes for 6G enabled NTN-assisted IoT technologies: recent developments, prospects and challenges," *IEEE Internet of Things Magazine*, vol. 7, no. 1, pp. 48-54, 2024. <https://doi.org/10.1109/IOTM.001.2300234>
- [41] K. K. Li, X. Gao, and X. G. Xia, "Channel estimation for LEO satellite massive MIMO OFDM communications," *IEEE Transactions on Wireless Communications*, vol. 22, no. 11, pp. 7537-7550, 2023. <https://doi.org/10.1109/TWC.2023.3252895>
- [42] S. Dey, T. Bharti, and R. Budhiraja, "RSMA in massive MIMO systems: analysis and optimization in correlated Rician-faded channels with phase shifts and aging," *IEEE Transactions on Communications*, vol. 72, no. 1, pp. 239-256, 2024. <https://doi.org/10.1109/TCOMM.2023.3299968>
- [43] J. Huang, Y. Yang, L. Yin, D. He, and Q. Yan, "Deep reinforcement learning-based power allocation for rate-splitting multiple access in 6G LEO satellite communication system," *IEEE Wireless Communications Letters*, vol. 11, no. 10, pp. 2185-2189, 2022. <https://doi.org/10.1109/LWC.2022.3196408>
- [44] S. Mahboob and L. Liu, "Revolutionizing future connectivity: a contemporary survey on AI-empowered satellite-based non-terrestrial networks in 6G," *IEEE Communications Surveys & Tutorials*, vol. 26, no. 2, pp. 1279-1321, 2024. <https://doi.org/10.1109/COMST.2023.3347145>
- [45] C. Qi, J. Wang, L. Lyu, L. Tan, J. Zhang, and G. Y. Li, "Key issues in wireless transmission for NTN-assisted Internet of Things," *IEEE Internet of Things Magazine*, vol. 7, no. 1, pp. 40-46, 2024. <https://doi.org/10.1109/IOTM.001.2300206>
- [46] Z. Zhang, Y. Chi, Q. Guo, Y. Li, G. Song, and C. Huang, "Asynchronous grant-free random access: receiver design with partially uni-directional message passing and interference suppression analysis," *IEEE Internet of Things Journal*, vol. 11, no. 9, pp. 15416-15433, 2024. <https://doi.org/10.1109/JIOT.2023.3347657>
- [47] Z. Ding, R. Schober, and H. V. Poor, "Impact of NOMA on age of information: a grant-free transmission perspective," *IEEE Transactions on Wireless Communications*, vol. 23, no. 5, pp. 3975-3989, 2024. <https://doi.org/10.1109/TWC.2023.3313612>

- [48] Y. Ma, G. Ma, B. Ai, J. Liu, N. Wang, and Z. Zhong, "OTFCS-modulated waveform design for joint grant-free random access and positioning in C-V2X," *IEEE Journal on Selected Areas in Communications*, vol. 42, no. 1, pp. 103-119, 2024. <https://doi.org/10.1109/JSAC.2023.3322796>
- [49] M. Harounabadi and T. Heyn, "Toward integration of 6G-NTN to terrestrial mobile networks: research and standardization aspects," *IEEE Wireless Communications*, vol. 30, no. 6, pp. 20-26, 2023. <https://doi.org/10.1109/MWC.005.2300207>
- [50] 3GPP, "Study on new radio (NR) to support non-terrestrial networks (Release 15)," Technical Report #38.811, 2018 [Online]. Available from: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3234>.
- [51] 3GPP, "Study on using satellite access in 5G (Release 16)," Technical Report #22.822, 2018 [Online]. Available from: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3372>.
- [52] 3GPP, "Study on supporting unmanned aerial systems (UAS) connectivity, identification and tracking (Release 17)," Technical Report #23.754, 2021 [Online]. Available from: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3575>.
- [53] 3GPP, "Study on management and orchestration aspects of integrated satellite components in a 5G network (Release 17)," Technical Report #28.808, 2021 [Online]. Available from: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3617>.
- [54] 3GPP, "Study on application layer support for unmanned aerial systems (UAS) (Release 17)," Technical Report #23.755, 2021 [Online]. Available from: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3588>.
- [55] 3GPP, "Support of uncrewed aerial systems (UAS) connectivity, identification and tracking; stage 2 (Release 17)," Technical Specification #23.256, 2021 [Online]. Available from: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3853>.
- [56] 3GPP, "Study on PLMN selection for satellite access (Release 17)," Technical Report #24.821, 2021 [Online]. Available from: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3811>.
- [57] 3GPP, "Application layer support for uncrewed aerial system (UAS); functional architecture and information flows (Release 17)," Technical Specification #23.255, 2021 [Online]. Available from: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3843>.
- [58] 3GPP, "LTE IoT NTN operating bands (Release 18)," Technical Report #36.764, 2021 [Online]. Available from: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=4216>.
- [59] 3GPP, "Study on requirements and use cases for network verified UE location for non-terrestrial-networks (NTN) in NR (Release 18)," Technical Report #38.882, 2022 [Online]. Available from: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=4074>.
- [60] 3GPP, "Study on 5G system with satellite backhaul (Release 18)," Technical Report #23.700-27, 2022 [Online]. Available from: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=4000>.
- [61] 3GPP, "User equipment (UE) radio transmission and reception for satellite access (Release 18)," Technical Specification #36.102, 2023 [Online]. Available from: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=4063>.
- [62] 3GPP, "NG-RAN; architecture description (Release 18)," Technical Specification #38.401, 2020 [Online]. Available from: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3219>.
- [63] 3GPP, "Solutions for NR to support non-terrestrial networks (NTN): non-terrestrial networks (NTN) related RF and coexistence aspects (Release 18)," Technical Specification #38.863, 2024 [Online]. Available from: <https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3926>.
- [64] 3GPP, "Study on uncrewed aerial vehicle (UAV) phase 3 (Release 19)," Technical Report #22.843, 2023 [Online]. Available: https://www.3gpp.org/ftp/tsg_sa/WG1_Serv/TSGS1_104_Chicago/post_SA_draft_specs/z_22843-j20.docx.

- [65] 3GPP, "Study on satellite access - phase 3 (Release 19)," Technical Report #22.865, 2023 [Online]. Available:
<https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=4089>.
- [66] 3GPP, "Study on management aspects of NTN – phase 2 (Release 19)," Technical Report #28.874, 2024 [Online]. Available:
<https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=4268>.
- [67] 3GPP, "Study on management of IAB nodes (Release 19)," Technical Report #28.875, 2024 [Online]. Available:
<https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=4265>.
- [68] 3GPP, "Uncrewed aerial system (UAS) support in 3GPP (Release 19)," Technical Specification #22.125, 2022 [Online] Available:
<https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3545>.
- [69] 3GPP, "Service requirements for the 5G system (Release 18)," Technical Specification #22.261, 2024 [Online] Available:
<https://portal.3gpp.org/desktopmodules/Specifications/SpecificationDetails.aspx?specificationId=3107>.
- [70] A. Hamissi and A. Dhraief, "A survey on the unmanned aircraft system traffic management," *ACM Computing Surveys*, vol. 56, no. 3, article no. 68, 2023. <https://doi.org/10.1145/3617992>
- [71] J. Li, X. Xiong, Y. Yan, and Y. Yang, "A survey of indoor UAV obstacle avoidance research," *IEEE Access*, vol. 11, pp. 51861-51891, 2023. <https://doi.org/10.1109/ACCESS.2023.3262668>
- [72] X. Luo, H. H. Chen, and Q. Guo, "LEO/VLEO satellite communications in 6G and beyond networks—technologies, applications, and challenges," *IEEE Network*, vol. 38, no. 5, pp. 273-285, 2024. <https://doi.org/10.1109/MNET.2024.3353806>
- [73] A. A. Zaid, B. E. Y. Belmekki, and M. S. Alouini, "eVTOL communications and networking in UAM: requirements, key enablers, and challenges," *IEEE Communications Magazine*, vol. 61, no. 8, pp. 154-160, 2023. <https://doi.org/10.1109/MCOM.004.2300061>
- [74] H. Hojeij, M. Sharara, S. Hoteit, and V. Veque, "On flexible placement of O-CU and O-DU functionalities in Open-RAN architecture," *IEEE Transactions on Network and Service Management*, vol. 22, no. 1, pp. 660-574, 2024. <https://doi.org/10.1109/TNSM.2024.3476939>
- [75] Y. Shaengchart and T. Kraiwanit, "The SpaceX Starlink Satellite Project: business strategies and perspectives," *Corporate and Business Strategy Review*, vol. 5, no. 1, pp. 30-37, 2024. <https://doi.org/10.22495/cbsrv5i1art3>
- [76] N. A. Khan and S. Schmid, "AI-RAN in 6G networks: state-of-the-art and challenges," *IEEE Open Journal of the Communications Society*, vol. 5, pp. 294-311, 2023. <https://doi.org/10.1109/OJCOMS.2023.3343069>